

Deceived Bilateral Filter for Improving the Classification of Football Players from TV Broadcast

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Abstract—This paper presents a novel image abstraction technique, the deceived bilateral filter (DBF). The DBF combines border sharpening with simplification of homogeneous regions to achieve moderate de-blurring and colour emphasis. The aim of the DBF is to remove noise of colour and shape descriptors (histograms and contours) of the blobs corresponding to football players. The experiment implemented for this paper consists in an unsupervised histogram based classification of the blobs based on the team of the football players aiming to evaluate histogram de-noising. The results show an average classification success rate improvement of 11%.

Keywords—image analysis; deceived bilateral filter

I. INTRODUCTION

The explosion of the internet and fast communication technologies brought together with the increasing availability of more computationally efficient and cheaper systems is boosting the image and video processing fields, with an increasing number of algorithms and techniques with different objectives. Thus, the analysis of video streams and images coming from different sources with rising types of applications is becoming a strong trend. The analysis of human, animals and even micro-organism activities fall within the challenges of such fields, reaching interesting research results and more recently, producing commercial applications [1].

A very popular and also well studied human activity, is football. It is one of the sports with most fans around the world, generating millions of Megabytes of data (videos, statistics, and more) every day. The automatic generation of match and player performance summaries in order to solve different fans and coaches queries, is one of the most interesting applications within the field [2], [3]. An attempt to solve the problems present in such application, the ACE-Football system was devised. ACE-Football is a computational analysis framework of football digital videos, for the automated analysis of the game. This abstract model is represented by a multi-layer architecture [4]–[7]. Lower levels deal with video shots finding, player localisation and tracking, while higher levels aim to extract tactical and strategic game information. Figure 1 details the hierarchical architecture of ACE, with the logical components of lower layers represented at the upper part of the figure.

In order to improve the performance of the spatial segmentation, target detection and localisation modules, the deceived

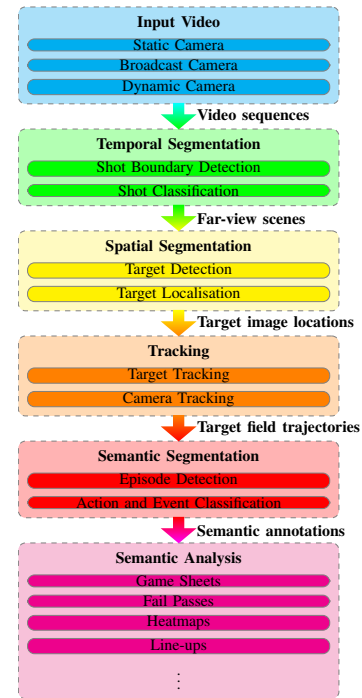


Fig. 1: Logical components in the layered architecture of ACE-Football

bilateral filter (DBF), a novel image abstraction algorithm, is presented in this work. The improvement will be obtained by means of motion blur de-noising and emphasizing of colours.

In section II a description of the current algorithms implemented in the target localisation and tracking components of the ACE-Football will be reviewed. The problem will be stated in section III, as well as the shortcomings of the current algorithms. In section IV, the previous work will be presented. The section V will be dedicated to the presentation of the proposed DBF. In section VI we devise an experiment to evaluate the results of the pre-processing with the DBF, to finally present the conclusions and future work in section VIII.

II. ACE-FOOTBALL: SPATIAL SEGMENTATION AND TRACKING

A. Target detection

The target localisation logical component in the layered architecture of ACE-Football implements a homogeneity-based segmentation algorithm, using the local spatial variance. Thus, the main assumption of this approach is the strong homogeneity of the football field, in spite of the candidate regions corresponding to the players blobs. The local spatial variance σ^2 is computed in the neighbourhood of every pixel p (with a pixel $p = (x, y)$ as an ordered pair) in the image at time t , $I_{t,var}(p)$.

Later, the empirical probability density of the local variance $h(k)_{t,var}$ is modelled as a bi-log-normal distribution, for which a threshold k_{th} is obtained using Otsu's maximum likelihood algorithm. The image is then thresholded using the value k_{th} to obtain the low-variance and high variance regions. The resulting regions are combined with the chromatic information of the dominant colour which describes the football field. After the aforementioned stages, morphological operations are performed in order to deal with spurious regions, to finally obtain the candidate regions $R_{t,c}$ corresponding to the blobs of the football players.

B. Target tracking

This stage aims to associate each candidate region $R_{t,c}$ in time t to the corresponding player region in time $t+1$, $R_{t+1,c}$. To solve such task the target tracking component implements an adaptive template matching [8] mixed with a bipartite graph matching.

Such approach represents the state of the system through a target state vector $X_{t,c}$ for each candidate region $R_{t,c}$. The target state vector includes descriptors as the shape of the silhouette, colour and texture information. The last two descriptors are represented in a colour histogram. The stacked vector corresponds to the set of the target vectors in time t , $\Gamma_t = \{X_{t,c} | c \in \{R_{t,c}\}\}$. The bipartite graph is then built as $B = (\Gamma_t, \Gamma_{t+1}, E, W)$ with $E = \{(X_{t,1}, X_{t+1,1}), \dots, (X_{t,n}, X_{t+1,m})\}$ representing the edges, and the weights by $W = (w_{1,1}, \dots, w_{m,n})$ which measures the match of the target descriptors between two edges. In order to eliminate implausible weights, a reduction is applied using a basic kinematic model of a player movement. This consists on defining a maximum target displacement between frames, discarding the connectivity between the edges of the graph corresponding to players that are far from each other.

The performance of the aforementioned tracking approach is highly dependent on the shape and colour descriptors of the candidate regions $R_{t,c}$ precision. Hence, noise caused by blurring or heavy lighting variations, affect the colour histogram and the contour of $R_{t,c}$. To deal with such noise, an algorithm to perform blurring de-noising and emphasize colour of the regions $R_{t,c}$ must be devised.



Fig. 2: Sample image of a football footage.



Fig. 3: A region $R_{t,c}$ with greenish pixels.

III. PROBLEM DEFINITION

Performing target localisation and tracking algorithms stated in section II using standard resolution videos (480p and 360p) as input, faces important shortcomings. A very frequent and important one is the image blurring caused by the players, camera movement and focus, along with poor image contrast and heavy lighting variations. As seen in the example figure 2 of an input image I_t , the background pixels values have a considerable presence in the segmented players blobs, as the motion blur, poor contrast, and the presence of shadows degrades the image. This phenomena worsens target detection and tracking stages performance. More specifically for the homogeneity based segmentation, as spurious pixels might be included in the candidate regions $R(p)_{t,c}$ as seen in the sample figure 3.

This causes an increased miss rate of the target localisation stage as the bipartite graph is prone to present imprecise target descriptors corresponding to the colour-texture and contour information of the blob. Noise pixels caused by motion blur also worsen the adaptive template matching algorithm performed in the target localisation stage.

Hence the blurring of the $R_{t,c}$ regions, consisting in the presence of aleatory pixels is the first problem to solve in this current work. Together with the de-noising task, emphasizing the perceptual characteristics of such regions $R_{t,c}$ regions (colour) is important in order to improve the performance of the aforementioned stages. The specific characteristics of the noise present are:

- Moderate blurring, caused by the camera focus, distance or low resolution.
- Blurring causes the appearance of greenish pixels inside the $R_{t,c}$ regions as in figure 2, which are considered as noise.
- Blurring parameters are unknown as we cannot suppose

a camera position, movement, focus or type.

- Variations of lighting caused by the presence of shadows is likely to affect histogram and contours of the candidate regions $R_{t,c}$.

Summarizing, the task of the algorithm is to attenuate the noise caused by blurring and improve the colour contrast and emphasis of the pixels p in the $R(p)_{t,c}$ regions with the aim of improving the performance of the target localisation and tracking components in ACE-Football.

IV. PREVIOUS WORK

This section provides a short survey of blurring removal, image sharpening, de-noising and image abstraction techniques found in the professional literature.

Blurring and contrast loss is modelled through a point spread filter (PSF) $h(p)$. It represents the system capability to diffuse light in a certain point. The degraded image $g(p)$ is defined as:

$$g = h * f + n \quad (1)$$

with f representing the original image, and n the additive noise.

Thus the equation 1 in the frequency domain is defined as:

$$G = F H + N. \quad (2)$$

Statistical based approaches aim to estimate the degrading kernel h in order to perform de-noising, while deterministic approaches assume fixed blurring qualities. Deterministic techniques can be divided in two types: models based on a surface interpretation of an image and transformation based models. The first type of techniques are formulated and performed based on a surface interpretation of the image, and the later performs a transformation of the image to at least formulate the operator.

A. Blur kernel estimation based techniques

The problem of estimating the blur kernel is also known as blind deconvolution. Algorithms within this category are usually more complex as a blurring kernel estimation stage is added. Blind deconvolution aims to estimate the reconstruction filter T which is able to obtain an approximation of the image $\hat{F}$:

$$\hat{F} = G T. \quad (3)$$

The reconstruction filter T can be either estimated given the input image or a series of images, or fixed, assuming a degradation function [9]. For instance, in [10] a Bayesian approach is implemented in order to estimate the blur kernel in images taken from a shaky camera. The following are papers which proposed different blind deconvolution techniques [11]–[14]. In general, aforementioned blind deconvolution variants aim to eliminate heavy motion blur degradations, with the need to invest computational resources to estimate the degradation through one or a series of images, with numerous iterations to achieve satisfactory results.

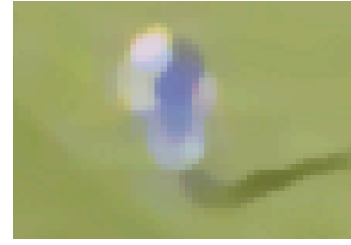


Fig. 4: Mazorra regularized implementation of a shock filter tested output with 35 iterations.

B. Fixed blur kernel techniques

Also known as non-blind deconvolution techniques, algorithms in this category assume fixed characteristics of the degradation function T in order to calculate the reconstructed image $\hat{F}$. The major paradigms of image processing, namely partial differential equations (PDE) and transformation based operators, present many techniques related to image de-blurring, sharpening and de-noising. Sub-sections IV-A, and IV-B present the principal concepts and previous work developed in such paradigms.

1) PDE based models: This framework is based on the interpretation of an image as a surface. Such interpretation allows the use of geometric tools including physical diffusion energy model in surfaces by variational calculus. In such model the energy propagates across the surface over time, which is reflected in the model by changing the image state through iterations.

The strength of this approach rests in its mathematical rigourousity and sensitivity to geometry. However, the common methods in this framework are usually poor to detect the overall trend of the signal, making them sensitive to noise [15]. Anisotropic diffusion as defined in [16] is a dissemination based technique.

Another operator used for blur removal and edge sharpening filter is the shock filter proposed in [17]. The shock filter iteratively improves edge slope applying morphological operations of dilation and erosion on the surface variations (edges). The original shock filter proposal has the disadvantage of amplifying noise. To alleviate this problem, many variations have been proposed including several regularization methods as presented in [18], [19]. However, such approaches present a high computational cost, in addition to the strong dependence to local variations still might cause noise amplification. In figure 4 the shock filter implementation proposed in [18] is used with 35 iterations, and it can be seen that the edges are steeper, however colours are not emphasized enough as only borders are modified. This leads to a modest improvement of histogram de-noising in cases of motion blur or strong shadow presence. For more details about the shock filter see [20].

2) Transformation based models: Models within this framework are based in the transformation of the image in the spatial domain g to a certain domain using a set of base functions or probabilistic approximations (Fourier, Wavelets, Histogram, etc). The transformed image $\bar{G}$ is used to either ap-

ply the reconstruction operator T in such domain or just design it to later apply it back in the spatial domain. Most of the deblurring methods within this framework work well in moderate degradation caused by motion blur, as such techniques handles defocus blurs and very moderate motion blur reasonably well [9]. Such features make this framework interesting for the problem at hand. The following are techniques based on transformation models.

a) Histogram equalization: Several histogram equalization based techniques aim to enhance the contrast in an image. These techniques perform a transformation to the histogram of the entire image or a region. More recent adaptive approaches as adaptive histogram equalization (AHE) [21] and the contrast limited adaptive histogram equalization (CLAHE) presented in [22], implement a region based histogram equalization. However, adaptivity is limited to relatively large regions as a valid histogram precises a considerable quantity of pixels, and colour emphasis might be limited.

b) Unsharped masking: A popular approach for deblurring is unsharped masking (USM) [9], which relies on the concept of amplifying the high frequency components. Such task can be performed either in the spatial or frequency domains. A basic approach to emphasize high frequency components is adding the degraded image g the high-pass filtered term, usually with a Laplacian filter. This operation results in a border emphasis. Equation 4 states the aforementioned concept mathematically.

$$\hat{f}_{UM} = g + \lambda z \quad (4)$$

$$z = l \square g \quad (5)$$

whereas z corresponds to the Laplacian filtered image, λ is the gain coefficient, and in Equation 5 l is defined as in [23].

$$l = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix} \quad (6)$$

The Laplacian filter is also used as border detector, as z is an approximation of the second derivative of g . Thus the

image reconstruction equation 4 obtains an image with the borders emphasized as the second derivative presents an over-shoot and under-shoot in the edges of the image (from now on

over-shoot and under-shoot will be referred just as over-shoot). This is illustrated in figure 5.

Over-shooting, brought together with the use of the mask defined in the equation 6, which accounts for the closest neighbouring pixels, produces a very often undesired ringing effect. The energy concentration in such peaks presented as over-shooting is not treated by the authors in [23]. However the authors propose a solution to another common problem in the USM based approaches, which is the amplification of noise in $\hat{f}$. Such solution consists on adjusting the gain λ in a local basis, using the local variance. A similar proposal is found in [24], with an adaptive sharpening algorithm that

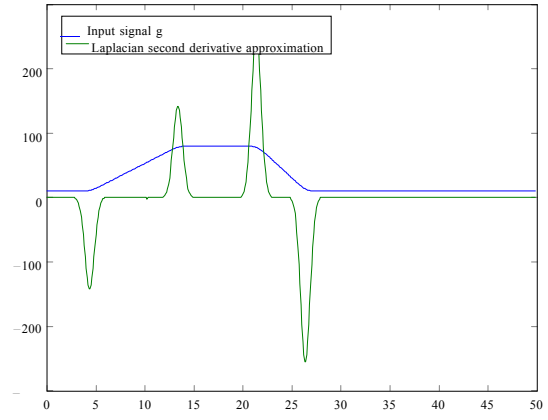


Fig. 5: Laplacian approximation of the second derivative, used by the USM sharpening method.

varies a λ trained for different types of regions. In this approach, λ can be negative, which makes it capable of removing noise.

Another adaptive sharpening algorithm is proposed in [25], which consists of a low pass filter and a high-pass filter locally adapted in order to improve edge acuity. The filter coefficients are updated recursively. Such approach still presents the ringing effect in the resulting image.

To improve the slope without generating such over-shooting, the authors in [26] proposed an algorithm for edge enhancement. This algorithm detects the direction of the edges to latter apply 1-D operator in order to improve the edges. However, the authors pointed the presence artifacts in the corners and little improvement in circular regions.

c) Bilateral filter based sharpening methods: The bilateral filter was proposed in [27]. The core concept of the bilateral filter consists in weighing both the geometric and range domain proximity for each pixel, using a Gaussian kernel in both cases. Such operation leads to the noise removal in homogeneous regions while preserving the edges. The mathematical definition of the bilateral filter is given in equation 7.

$$\hat{f}_{BL}(p) = \left(\sum_m h(p, m) \right)^{-1} \left(\sum_m h(p, m) g(m) \right) \quad (7)$$

where $h(p, m)$ corresponds to the bilateral filter kernel:

$$h(p_0, m) = \exp \left(-\frac{(m - p_0)^2}{2\sigma_s^2} \right) \exp \left(-\frac{(g(m) - g(p_0))^2}{2\sigma_r^2} \right) \quad (8)$$

The pixel m iterates through a window W with dimensions $N_1 \times M_1$. The spatial and range deviations are denoted by σ_s and σ_r respectively. The second exponential term in the equation 8 weights the pixel value closeness, allowing edge preservation. It is worth to highlight that larger values of the three parameters leads to a blurrier image, with major

influence of the range deviation σ_r which defines edge preservation.

Several researchers have connected to bilateral filter noise removal operations with other techniques based on different paradigms, such as diffusion based methods [28].

In [29] the authors modified the bilateral filter for noise removal and edge enhancement by adding a high-pass filter. The algorithm generates artifacts and ringing as stated in [29].

The adaptive bilateral filter (ABF) proposed in [30] is based in the bilateral filter as proposed in [27]. In the ABF, the slope of the edges are improved, thus attenuating blur noise. However the colours are not emphasized, acting similarly to the shock filter tested in figure 4.

C. Image abstraction techniques

The approaches presented in sub-sections IV-A IV-B can be referred as low level image processing algorithms. Such techniques aim to modify low level properties of the images like contrast, edges strength and noise presence. Image abstraction approaches as the ones found in [31]–[34] intend to simplify the visual content of images and also emphasize perceptually important features as colours and edges [31]. Image abstraction techniques are frequently used as a tool to generate artificial images with artistic purposes.

For the task at hand in this paper, image abstraction can serve to simplify and emphasize image's colour content. As presented in [31], the evaluation of the proposed abstraction framework through a user-study found that participants were able identify abstracted faces of known persons faster, when compared to the identification with original photographs. The authors implemented an image abstraction algorithm using several iterations of the bilateral filter to later apply a colour quantization with the aim to simplify low contrast regions.

More recent works focus on the edge detection and geometrical smoothing as seen in [33]. However, an image abstraction filter which emphasizes with more strength colour features is missing for our purpose in this paper.

V. PROPOSED SOLUTION

As stated in section III, the main purpose of our image abstraction algorithm is to emphasize colour and perform motion blur de-noising and image contrast improvement in small blobs corresponding to the players in a soccer match. From the techniques studied in section IV, we focus on the ones with a fixed kernel blur, as the motion blur problem in typical football matches videos is usually slight, caused by rapid movements of the camera and focus distance.

Our proposed solution is inspired in the usage of the bilateral filter as an image abstraction algorithm as in [31] as it proved to effectively simplify texture content. However, the use of an alternative bilateral based approach as in [29], [30] to improve image contrast and sharpness is necessary to emphasize colours. The use of a high-pass amplification based approach (USM) as in [29] presents the advantage of the possible use of an improved adaptivity of the high-pass filter to the signal, in spite of an histogram based approach as

in [30]. Also the use of a high-pass amplification for contrast enhancement is easier to control, and is able to provide further colour emphasis and not only steeper borders, when compared to a histogram based approach. However, as presented in the short survey in section IV, USM based methods cause artifacts as ringing.

A. The deceived bilateral filter (DBF)

We propose the combination of the bilateral filtering with USM as in [29], in order to achieve image sharpening and texture de-noising. Additionally, using an adaptive USM approach as the one proposed in [23] might diminish noise amplification. However, ringing is still a problem. Hence, we propose the combination of USM and the bilateral filter in a novel manner, with the aim to reduce the overshoot in the resulting signal of the USM $\hat{f}_{UM}$. The modification consists in the weighting of the range pixel distance using the input image g , instead of using for range domain weighting the image $\hat{f}_{UM}$. However, the new pixel values present in the edges of $\hat{f}_{UM}$ are used to calculate the output pixel values, thus distributing the energy present in the over-shoots. This produces a stronger overshooting attenuation in $\hat{f}_{UM}$, diminishing the ringing effect significantly. This concept is mathematically formalized as:

$$\hat{f}_{DBL}(p) = \left(\sum_m h(p, m) \right)^{-1} \left(\sum_m h(p, m) \hat{f}_{UM}(m) \right) \quad (9)$$

Where the kernel of the bilateral filter remains the same as in the original proposal, as seen in equation 8:

$$h(p_0, m) = \exp \left(-\frac{(m - p_0)^2}{2\sigma_s^2} \right) \exp \left(-\frac{(g(m) - g(p_0))^2}{2\sigma_r^2} \right) \quad (10)$$

In figure 7, an example signal is processed with the bilateral filter without deceiving it i.e, using as input of the original bilateral filter the USM output signal, and it is compared with the output of the DBF. The overshooting present in the sample signal of figure 5 is translated in the image domain as a ringing effect. Such effect causes imprecise region detection failures of homogeneity based segmentation algorithms as the one defined in sub-section II-A. Figure 6 shows how the attenuation of over-shooting in the synthetic test translates to a ringing effect diminution, when compared with a serial processing with the bilateral filter and USM. The over-shoots and under-shoots seen in figure 7 are almost eliminated for larger regions and suppressed in small ones. Also, figure 6 shows how contrast and colour emphasis is improved from the original image, using the same Laplacian gain λ . The proposed method takes a step further in the bilateral filter combination with the USM based edge sharpening method proposed in [35] as it effectively diminishes the ringing effect, and suppresses it in small regions.

For colour images, Tomasi and Manduchi recommend implementing the bilateral filter in a perceptually linear colour space such as CIE-Lab [27]. We implemented the DBF on the



Fig. 6: From left to right, the original input image, the bilateral filter plus the USM output and the DBF result.

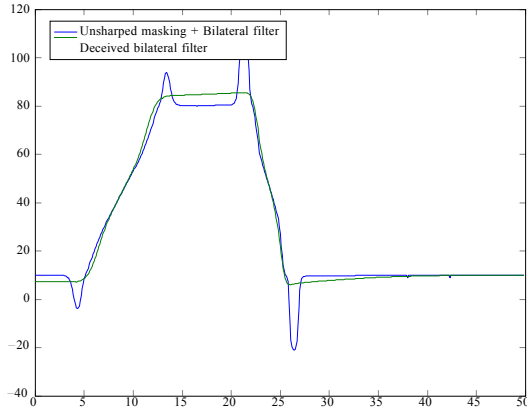


Fig. 7: DBF and serial bilateral filtering plus USM output comparison in an artificial signal.

three layers (L, A, B) combined, using a perceptual weighting as recommended by Tomasi and Manduchi.

VI. EXPERIMENT DEFINITION

As stated in section III, the colour and shape descriptors of a candidate region $R_{t,c}$, defined as the colour histogram and the region contours (relative positions of the region border from its centroid), are sensible to blur noise in the image. Motion blur caused by camera or player's movement and variable lighting in low resolution videos (240p or 360p, as in higher resolution videos motion blur noise presence is weaker) makes colour and shape descriptors less representative, as background colour information affects $R_{t,c}$. Thus, we expect that the DBF applied in the images of a soccer footage de-noise the colour histogram $h_{t,c}$ of a region $R_{t,c}$ and improve segmentation accuracy. For this paper, we designed an experiment to test the significance improvement of the colour descriptor. We chose to focus on the expected effect of histogram de-noising and colour emphasis of using the DBF for preprocessing. The experiment aims to evaluate the consistency of the colour histogram for all the blobs in a defined sequence of frames, performing unsupervised clustering in a batch fashion.

The experimental setting consists in performing an unsupervised classification of the players regions in a sequence of 6.2 seconds (with the k-means algorithm), with two classes,

each one corresponding to a team. The selected sequence presents strong shadows in the field as also motion blur noise (in this particular scene the players of Korea move very fast in an offensive action). The sequence is stored with a 360p resolution. The ground truth used to validate the classification results for the experiment was generated manually using the software tool Sensarea [36].

The steps of the experiment performed for this paper are detailed as follows:

- 1) Apply to a segment of a soccer video (Korea vs Greece, FIFA World Cup 2010) of 6.2 seconds V_{test} , with 25 frames per second the DBF, to obtain $V_{DBF,test}$.
- 2) The ground truth masks for each team M_{GT} are used to generate the normalized histograms $h_{t,c}$ corresponding to every region $R_{t,c}$. The calculation of the histograms $h_{t,c}$, is performed for the video sequences V_{test} and $V_{DBF,test}$, with the histograms of all the frames in the sequence stored in H_{test} and $H_{DBF,test}$. The total number of player blobs or samples is $C = 1185$, which corresponds to the total number of histograms used to perform the unsupervised clustering.
- 3) Using each data collection, H_{test} and $H_{DBF,test}$ as input for the k-means algorithm, a classification is performed, storing the cluster ownership for each data collection in R_{test} and $R_{DBF,test}$.

sequence R_{test} and $R_{DBF,test}$ is compared against the ground truth to obtain the overall success classification rate in each video sequence.

In order to measure the distance between two histograms, we implemented the Bhattacharyya distance as used in [7]. The histograms were processed in the three most commonly used colour spaces: HSV, CIE-LAB and RGB. The main goal in an unsupervised classification problem is to reach the most accurate classification using representation vectors with the lowest possible dimensionality, to reduce computational cost. Defining the vector with the most representative components is the main challenge in classification problems. Hence, we propose to perform the unsupervised classification test in different dimensionality spaces.

Intuitively, vector representation in the case for colour based classification of players must represent chromatic information, excluding illumination data, as players must be correctly classified independently from the illumination circumstances. Hence, we evaluate the unsupervised classification in the three colour models as follows. For the RGB-colour model R, G and B layers, are used to build the histogram. Such layers contain illumination, saturation and chromatic information, presenting the highest dimensionality.

We expect that the DBF improves clustering results as the colour are emphasized, minimizing the effects of motion blur and variable illumination. To evaluate the DBF impact on the saturation and chrominance qualities of the image, we performed a histogram clustering using the A and B layers. Finally, the lowest dimensionality representation evaluated is built using the H layer from the HSV colour space. In the

TABLE I: K-means classification success rate (%) , with and without the use of the DBF as a preprocessing stage using the RGB-colour space.

Video	B = 10 ³	B = 24 ³	B = 32 ³	B = 64 ³
Without DBF	60.02	60.02	60.02	60.04
With DBF	89.11	89.79	90.04	97.38
Improvement	+29.08	+29.76	+30.02	+37.34

TABLE II: K-means classification success rate (%) , with and without the use of the DBF as a preprocessing stage using the CIE-LAB-colour space.

Video	B = 10 ²	B = 24 ²	B = 32 ²	B = 64 ²
Without DBF	91.73	91.64	91.39	91.56
With DBF	92.65	92.23	91.98	90.80
Improvement	+0.92	+0.59	+0.59	-0.76

tests for the colour spaces where more than one layer is used, different binning numbers were tested $B = 10, 24, 32, 64$, resulting in a sample dimensionality $D = 10^L, 24^L, 32^L, 64^L$, where L stands for the number of layers.

We performed the DBF with two iterations, using a descending normalized Laplacian gain of $\lambda = 2$ for the first iteration and $\lambda = 1$ for the last one. The Laplacian mask for the USM implementation presents a dimension of 11×11 . As for the DBF parameters, the spatial and range deviations were valued as $\sigma_s = 7$ and $\sigma_r = 5$ and the window size in pixels is assigned as $W = 11$.

VII. RESULTS

For the RGB-colour space, the clustering results are shown in table I, using the defined binning schemes in section VI.

As table I shows, in the RGB-colour model based histograms, the use of the DBF brings an average of 31% classification success rate improvement.

For the CIE-LAB-colour space, the results obtained for each binning scheme are shown in table II. In this case, a modest success rate improvement is obtained after the use of the DBF, averaging a 0.33 % of success rate improvement.

Table III presents the results of performing the experiment stated in section VI, for the HSV colour space. In this case the number of bins used to build the histogram is $B = 256$.

The result improvement of performing an unsupervised classification using the hue layer is boosted to around 3% for the tested sequence.

TABLE III: K-means classification success rate (%), with and without the use of the DBF as a preprocessing stage using the HSV-colour space.

Video	B = 256
Without DBF	95.27
With DBF	98.22

VIII. CONCLUSIONS AND FUTURE WORK

The results shown in table I demonstrates an important success rate boost (averaging an 11(%) success rate boost across the tested color spaces), as the colour emphasis perform by the DBF improves the significance of the colour histogram under variable lighting and motion blur conditions. The results for the rest of the experiment averages lower success rate improvements, as the layers of the CIE-LAB, and HSV-colour spaces contain more significant chromatic information in spite with illumination data.

The highest success rate was achieved using the DBF and building the histograms with the hue layer in the HSV-colour space, with a 98.22% of success rate. With the exception of a minor decrease of the success rate using the A and B layers with 64 bins, the use of the DBF shows a consistent success rate improvement. It is worth to mention that such improvement rests entirely on the motion blur de-noising and colour emphasis, as the manual segmentation of the masks for the video sequences V_{test} and $V_{\text{DBF, test}}$ remained the same for both of them. As for the specific unsupervised colour classification task, we recommend to use a colour-spatial representation as the one proposed in [37]. The use of the DBF in a colour-spatial based classification could boost performance, however this must be explored.

The segmentation accuracy evaluation is pending (using the algorithm defined in sub-section II-A), but also a colour descriptor de-noising and significance evaluation with the automatic segmentation is valid as the segmented regions are prone to change with the use of the DBF. We expect a major improvement with the use of the automatically generated region blobs of the colour histogram significance, but also its evaluation is still pending. Also, an evaluation of the tracking algorithm defined in sub-section II-B is unsettled.

The DBF can be improved to make it less sensible to noise using concepts as an adaptive USM like the one proposed in [23] or using a variant of the bilateral filter based in rank order filtering or other techniques as in [38]–[40]. Performance wise, the DBF can be considered to be computationally expensive, specially when large windows are used. Also the computational cost of performing Gaussian weighting is important, and is treated in detail in [41]. We consider that the underlying concept of the DBF allows the implementation of performance optimized modifications of the bilateral filter, but this must be explored with more detail.

Finally, as short term future work, we aim to use the DBF in different applications, as the segmentation improvement of cancer cells. The qualities of the DBF makes it interest to use in general border sharpening and region de-noising problems, thus the interest of evaluating its use in different applications.

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